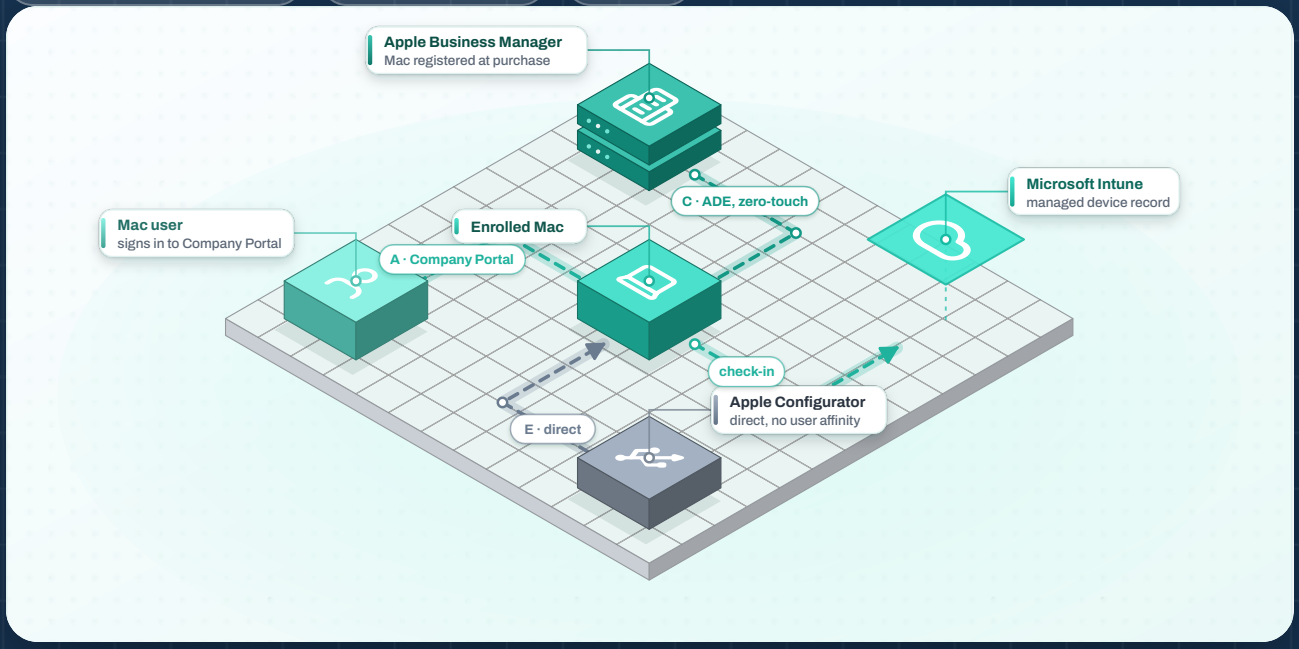
STANDARD OPERATING PROCEDURE · FIELD GUIDE SERIES

**SOP-EM-002**

# Enrolling macOS Devices into Microsoft Intune

Company Portal self-enrollment, Automated Device Enrollment through Apple Business Manager, direct enrollment, and the admin prerequisites behind them, with a Graph PowerShell readiness check.

Endpoint Management · Microsoft Intune · macOS




| | |
|---|---|
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### LEARNING OBJECTIVES

After working through this procedure you will be able to:

- Explain what an MDM enrollment profile, APNs, supervision and user-approved MDM are.
- Compare Company Portal, Automated Device Enrollment and direct enrollment for a given Mac.

- Walk a user through Company Portal enrollment and approve the management profile.
- Create and renew the APNs certificate and set enrollment restrictions in Intune.
- Connect Apple Business Manager to Intune with an ADE token and enrollment profile.
- Audit enrollment readiness (APNs, ADE tokens, restrictions, devices) with Graph PowerShell.

| TIME | DIFFICULTY | MINIMUM ROLE | RISK |
| --- | --- | --- | --- |
| ≈ 10 min per Mac (user); 60 min one-time admin setup | Beginner (user) / Intermediate (admin) | Intune Administrator (APNs); Enrollment programs role (ADE) | Low; an expired APNs cert breaks all Apple management |

CHAPTER 1

# Concepts You Need First

"Enrolling" a Mac means introducing it to a management server and getting the Mac to agree, in a way Apple considers trustworthy, that the server may configure it from now on. Everything an organization later does to that Mac (Wi-Fi profiles, compliance checks, Defender, software updates, a remote wipe) depends on this one relationship. If enrollment is done the wrong way, some of those capabilities are simply unavailable, and fixing it later usually means re-enrolling or even erasing the Mac. That is why this SOP spends a chapter on concepts before anyone clicks **Begin**.

> **KEY CONCEPT · A LEASE, NOT A SALE**
>
> Think of enrollment as signing a lease. The Mac (the tenant) agrees that the landlord (Intune) may inspect, configure and, if needed, reclaim the property. A *Company Portal* enrollment is a lease the tenant signs after moving in: valid, but the tenant can tear it up. *Automated Device Enrollment* is a lease written into the property deed before the tenant ever arrives: the building ships pre-registered to the landlord and the lease cannot simply be torn up.

## 1.1 The MDM enrollment profile and how Intune talks to a Mac

Apple devices are managed through Apple's **MDM protocol**. Enrollment installs an **MDM enrollment profile** (a signed configuration profile) that contains the server address and an identity certificate. From then on the Mac's built-in `mdmclient` polls the server for commands. Intune never connects *into* a Mac. Instead it asks Apple to nudge the Mac:

1. Intune sends a request to the **Apple Push Notification service (APNs)**: "wake device X".
2. APNs delivers a tiny silent notification to the Mac over its persistent connection to Apple.
3. The Mac connects to Intune over HTTPS and collects any pending commands and profiles.

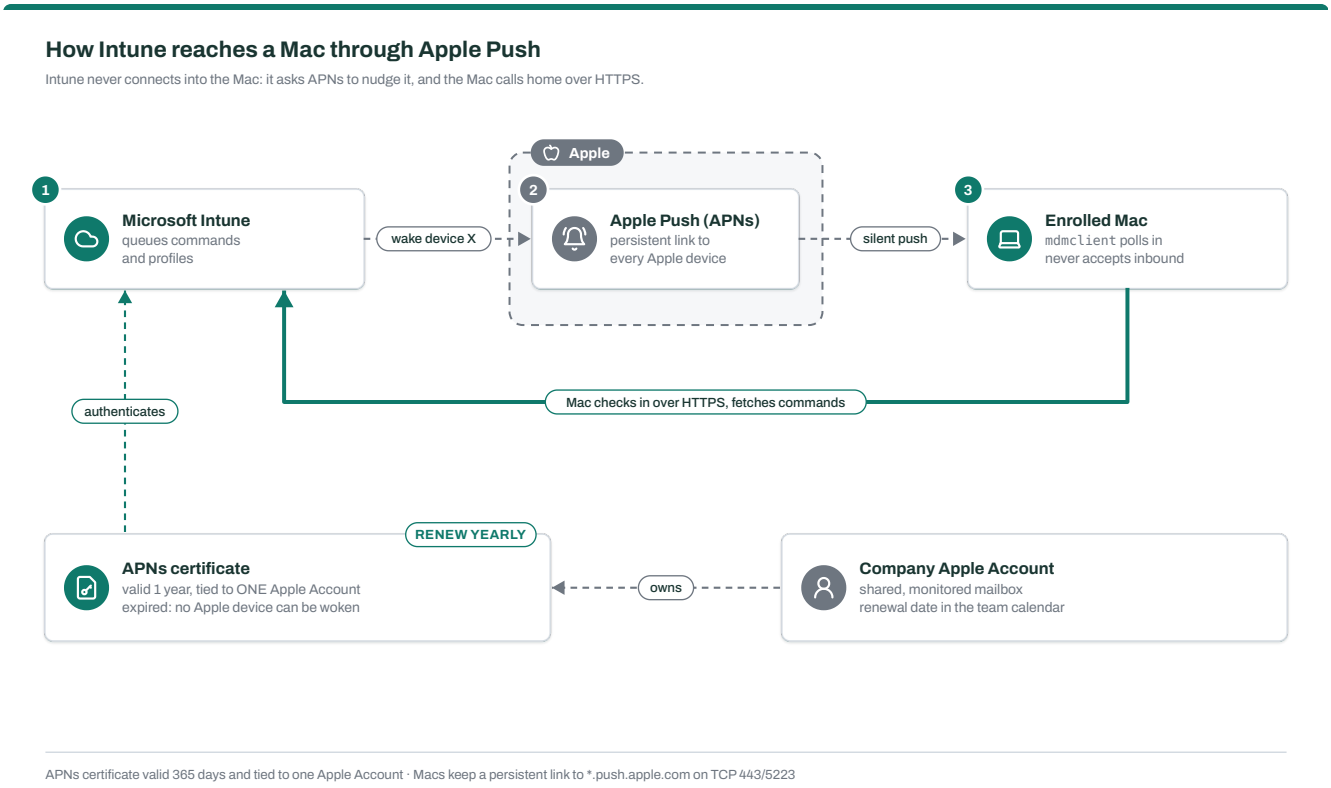



**Figure 1.** Because the Mac only ever calls out, the APNs certificate is the single point of failure: if it expires, Intune can no longer wake any Apple device.

**CAUTION · THE SINGLE POINT OF FAILURE**

The APNs certificate is valid for one year and is tied to the Apple Account that created it. Renewing it with a *different* Apple Account, or letting it expire and creating a new one, breaks the trust for every enrolled Mac, iPhone and iPad, which then have to re-enroll. Use a shared, monitored company Apple Account and put the renewal date in the team calendar.

## 1.2 User-approved MDM and supervision

Apple grants an MDM server different levels of power depending on how the device was enrolled:

| State | How a Mac gets it | What it unlocks |
|---|---|---|
| **User-approved MDM** | The user explicitly approves the profile in **System Settings › Device Management**, or the Mac enrolled through ADE. | Silent approval of system extensions, privacy (PPPC) payloads and kernel-level security settings. |

| | | |
|---|---|---|
| **Supervised** | Enrollment through Automated Device Enrollment (macOS 11 and later are supervised automatically). | Extra restrictions, managed software updates, Activation Lock bypass, bootstrap token escrow, non-removable MDM profile. |
| **Unsupervised, user-approved** | Company Portal or direct enrollment. | Most configuration and compliance; the user can remove management. |

## 1.3 Three ways to enroll

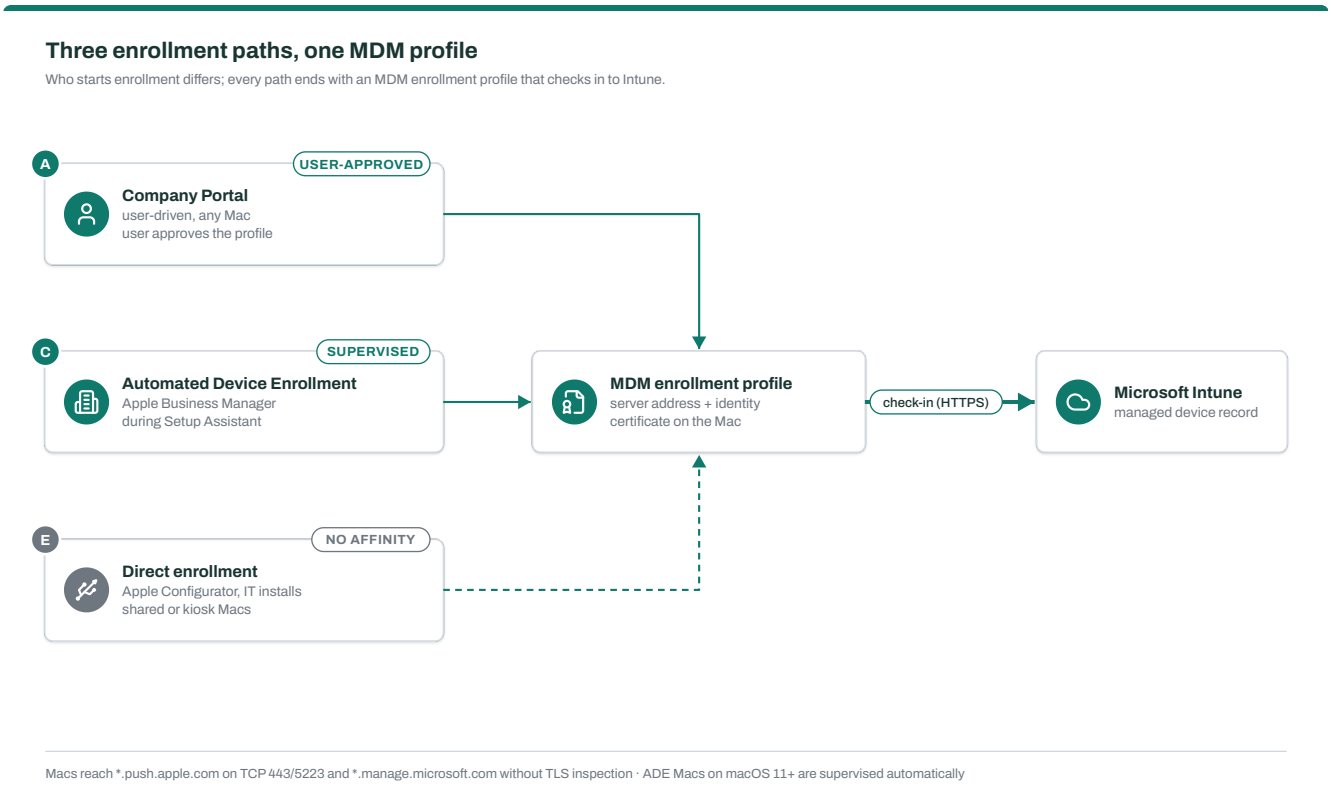



**Figure 2.** The paths differ in who starts enrollment and how tightly the Mac is bound to the organization, not in what lands on the Mac.

- **Company Portal (user-driven).** The user installs Microsoft's Company Portal app, signs in with a work account and approves the management profile. Works for any Mac, including ones bought at retail. The device gets *user affinity* (it belongs to the signed-in user), which conditional access and app deployment rely on.
- **Automated Device Enrollment (ADE).** Macs bought through Apple or an authorized reseller are registered to the organization in **Apple Business Manager (ABM)** (or Apple School Manager). ABM tells the Mac at first boot that it belongs to your Intune tenant, so enrollment is part of Setup Assistant. Supervised, locked and zero-touch.

- **Direct enrollment.** An administrator exports an enrollment profile from Intune and installs it on the Mac (for example with Apple Configurator). No user affinity; suited to shared or kiosk Macs where no user signs in to Company Portal.

**WHY IT MATTERS · OWNERSHIP**

Intune marks a Company Portal Mac as *Personal* by default unless its serial number was pre-registered as a **corporate device identifier**; ADE Macs are always *Corporate*. Ownership decides which policies apply and what IT may see or wipe, so plan it before the first enrollment, not after.

## 1.4 Choosing a method

**A · Company Portal (user)**

The user-facing walk-through, step by step with screenshots.

BEST FOR: EXISTING MACS, BYOD

**B · Admin prerequisites**

APNs certificate, enrollment restrictions, device limits, corporate identifiers, branding.

BEST FOR: EVERY TENANT, ONCE

**C · Automated Device Enrollment**

ABM + ADE token + enrollment profile; enrollment during Setup Assistant.

BEST FOR: NEW CORPORATE MACS

**D · Graph PowerShell**

Readiness audit: APNs and token expiry, restrictions, enrolled Macs; optional token sync.

BEST FOR: MONITORING, EVIDENCE

**E · Direct enrollment**

Exported enrollment profile installed by IT, no user affinity.

BEST FOR: SHARED / KIOSK MACS

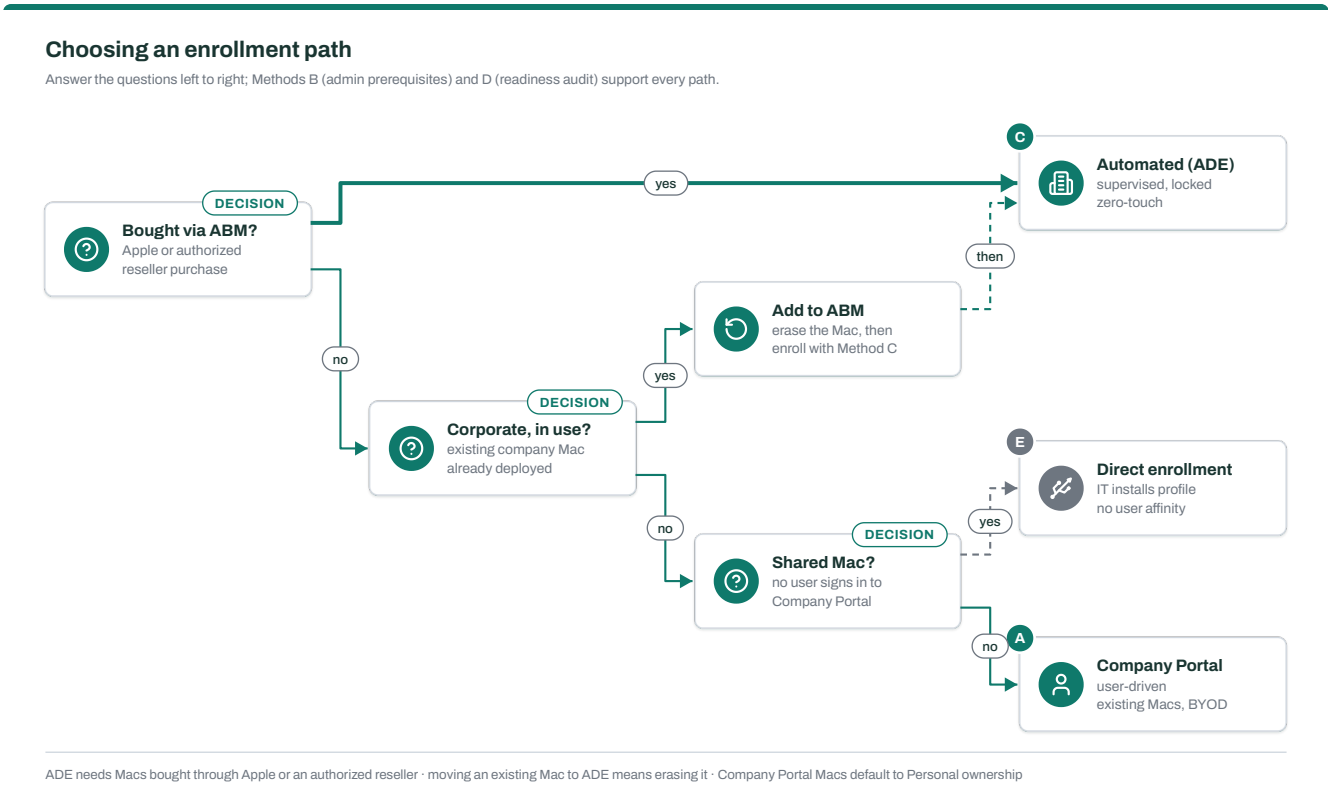



**Figure 3.** Purchase channel decides first: a Mac that came through Apple Business Manager should always take the supervised ADE path.

CHAPTER 2

# Access, Prerequisites and Planning

## 2.1 Level of access required

Enrollment spans three administrative systems: **Apple Business Manager** (device ownership), **Microsoft Entra ID** (tenant-level Intune roles) and **Intune RBAC** (enrollment settings). The end user also needs rights: an Intune license and, on the Mac, an administrator account to install software and approve profiles.

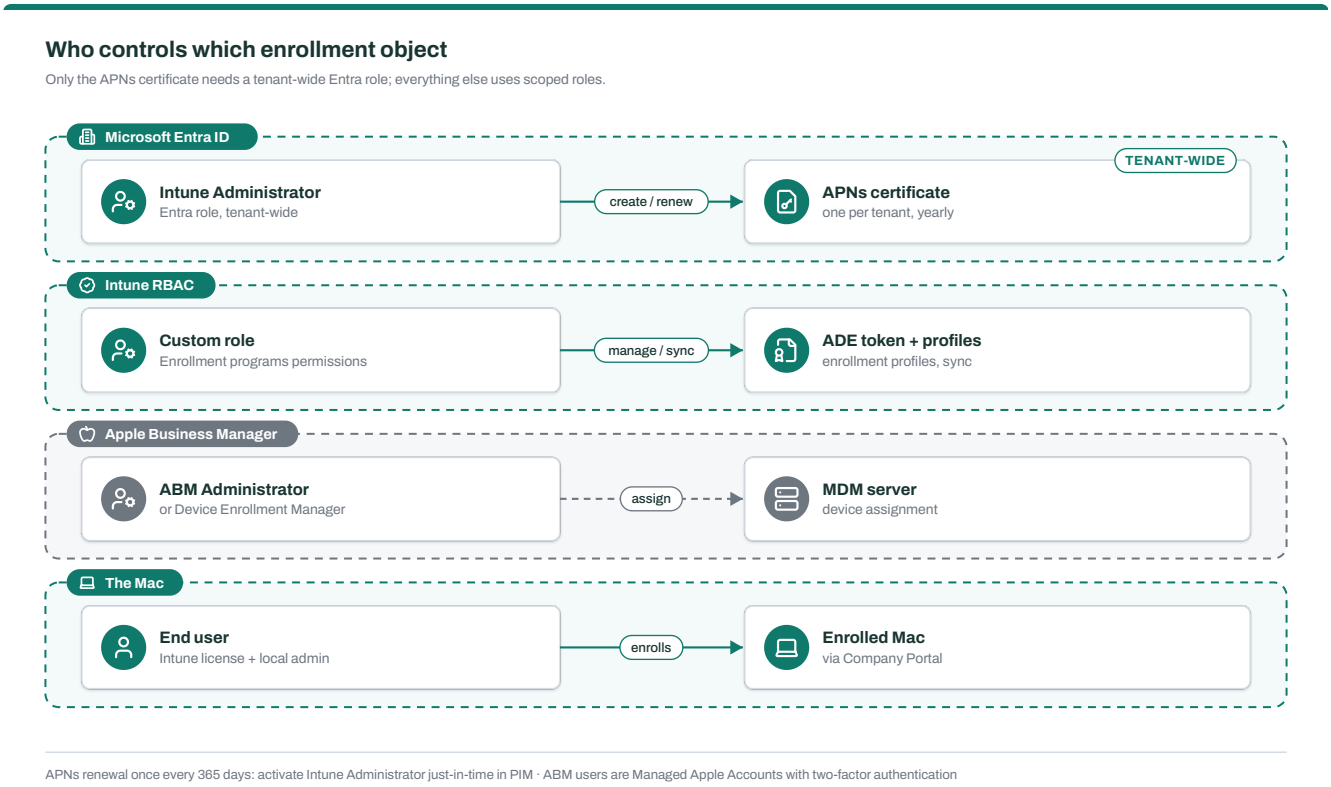



**Figure 4.** Keep the tenant-wide Intune Administrator role for the yearly APNs renewal and do day-to-day enrollment work with scoped roles.

| Task | Least-privilege role / permission | Notes |
|---|---|---|
| **Create or renew the APNs certificate** | Intune Administrator (Entra); Graph `DeviceManagementServiceConfig.ReadWrite.All` | Plus access to the shared company Apple Account. |
| **Add/renew ADE tokens, create enrollment profiles, sync devices** | Custom Intune role with *Enrollment programs* permissions | Intune Administrator also works but is broader. |

| | | |
|---|---|---|
| **Register Macs and MDM servers in ABM** | ABM *Administrator* or *Device Enrollment Manager* role | Managed Apple Account with two-factor authentication. |
| **Enrollment restrictions, device limits, corporate identifiers** | Intune Administrator, or a custom Intune role with the enrollment-restriction and corporate-identifier permissions | Changes affect every new enrollment; use change control. |
| **Company Portal branding** | Intune Administrator or custom role with *Organization › Update* | Sets the name users see ("Set up Contoso access"). |
| **Audit readiness by script** | Read Only Operator (Intune); Graph `DeviceManagementServiceConfig.Read.All`, `DeviceManagementManagedDevices.Read.All` | Read-only unless `-SyncAdeTokens` is used. |
| **Enroll a Mac (end user)** | Licensed Intune user; local administrator on the Mac | Standard users need IT to authenticate the install and the profile approval. |

**GOOD PRACTICE · JUST-IN-TIME PRIVILEGE**

The APNs renewal is the only task that needs Intune Administrator, and it happens once a year. Make that role *eligible* in Privileged Identity Management and activate it for the renewal window only. Day-to-day enrollment work uses the scoped custom role.

## 2.2 Prerequisites checklist

- [ ] Intune licenses assigned to users who will enroll Macs; MDM authority is Intune.
- [ ] Valid APNs certificate, more than 30 days from expiry (Chapter 4).
- [ ] macOS enrollment allowed by the platform restriction; personally owned Macs allowed or blocked deliberately.
- [ ] For ADE: an Apple Business Manager account verified for your organization and Macs purchased through Apple or a participating reseller (or added with Apple Configurator for iPhone).
- [ ] Macs can reach Apple (`*.push.apple.com`, TCP 443/5223) and Intune (`*.manage.microsoft.com`) without TLS inspection.
- [ ] Company Portal branding set and a support contact published for users.

## 2.3 Decisions to make up front

| Decision | Guidance |
|---|---|
| **Personal Macs** | Allow with limited policy, or block in the platform restriction. Blocking requires corporate identifiers or ADE for corporate Macs. |

| | |
|---|---|
| **User affinity (ADE)** | With user affinity for assigned Macs; without for shared/lab Macs. |
| **Locked enrollment** | Yes for corporate ADE Macs, so users cannot remove the MDM profile. |
| **Existing corporate Macs** | Company Portal now, or erase and re-enroll through ADE for supervision. |

CHAPTER 3

# Method A · Company Portal Enrollment (User Walk-through)

This chapter is written so it can be handed to a Mac user. It takes about ten minutes. The user needs their work email and password, their MFA method and the Mac's administrator password.

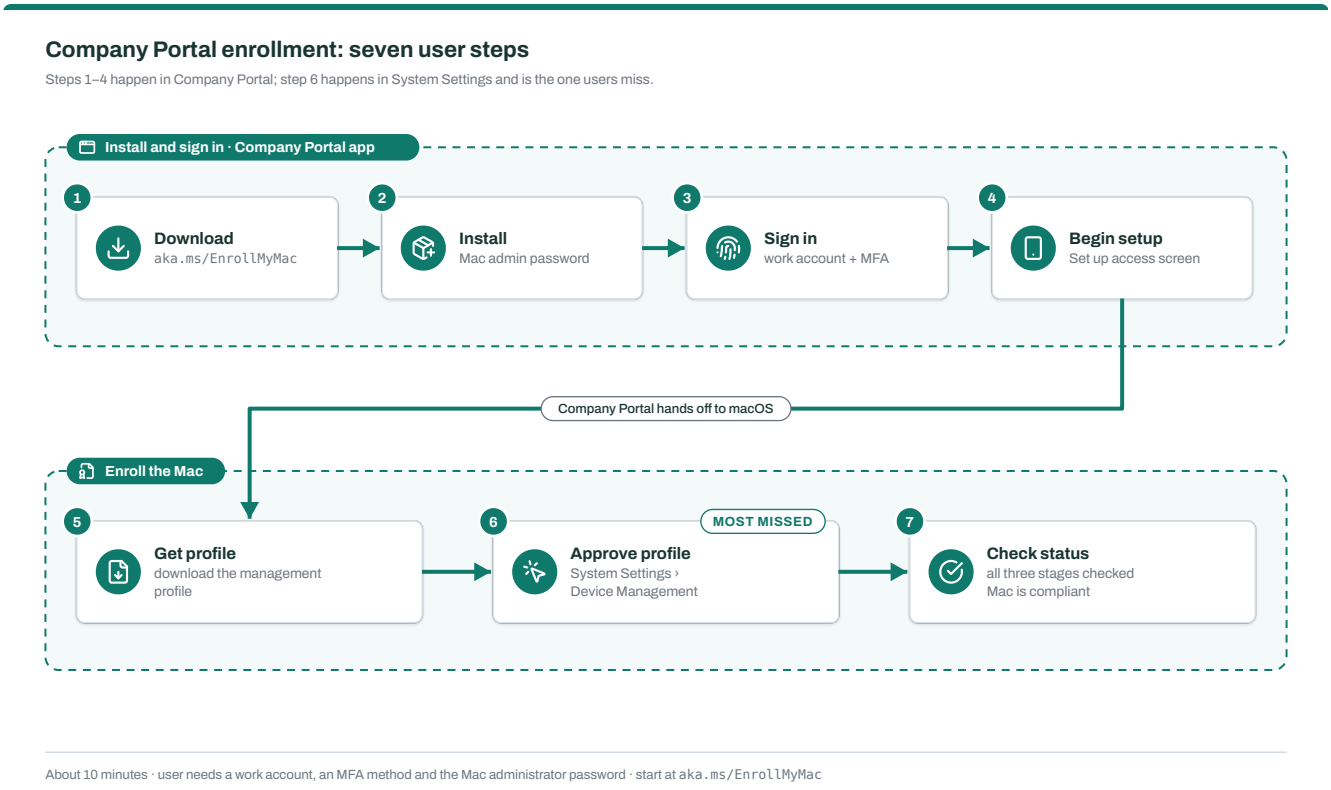



**Figure 5.** Warn users that step 6 leaves Company Portal for System Settings; skipping it is the most common reason a user-driven enrollment stalls.

## STEP 1–3 Install Company Portal

1. In Safari go to **aka.ms/EnrollMyMac**. The installer `CompanyPortal-Installer.pkg` downloads to the *Downloads* folder.
2. Double-click the package. On the welcome screen select **Continue**.
3. Continue through *License* and *Destination*, enter the Mac password when prompted, and wait for the installer to finish. On *The installation was successful* select **Close**.

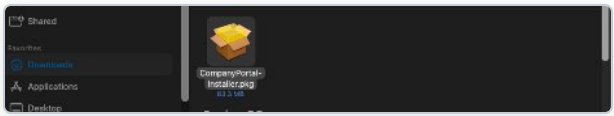



**Figure 6.** The downloaded installer package in Finder › Downloads.

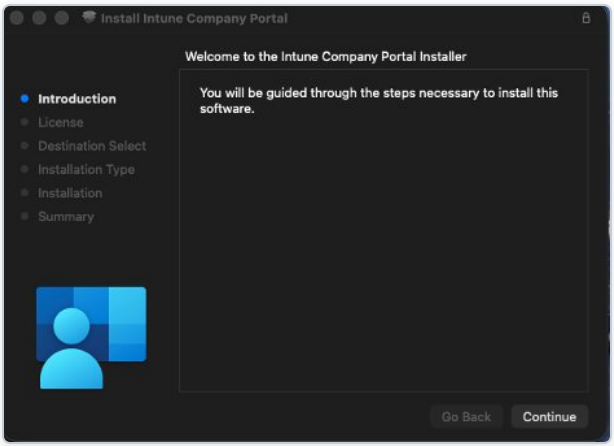



**Figure 7.** Installer welcome screen; select `Continue`.

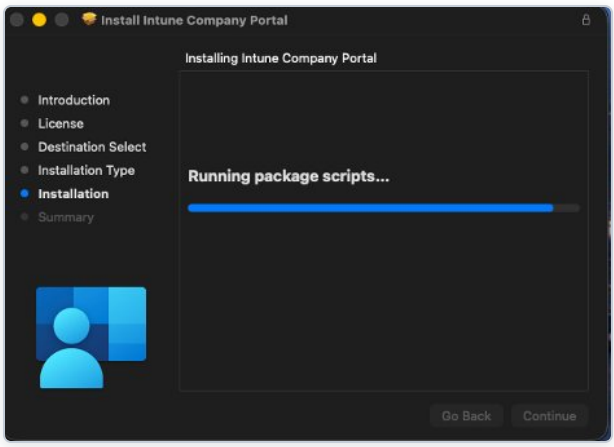



**Figure 8.** The installer runs its package scripts.

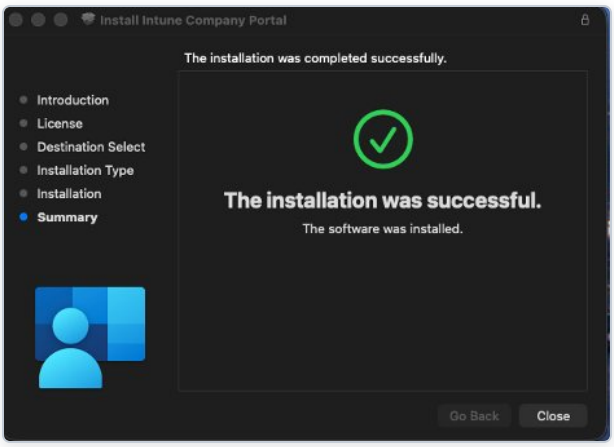



**Figure 9.** Installation successful; select `Close`.

## STEP 4–5 Sign in and begin setup

1. Open **Company Portal** from Applications or Launchpad, select `Sign in`, enter the work email and password and approve the MFA prompt.
2. On *Set up <organization> access* select `Begin`. Do not select `Postpone`: the device stays unenrolled and will not reach company resources.
3. Review the privacy information screen. It explains what the organization can and cannot see (it cannot see personal files, browsing history or messages) and select `Continue`.

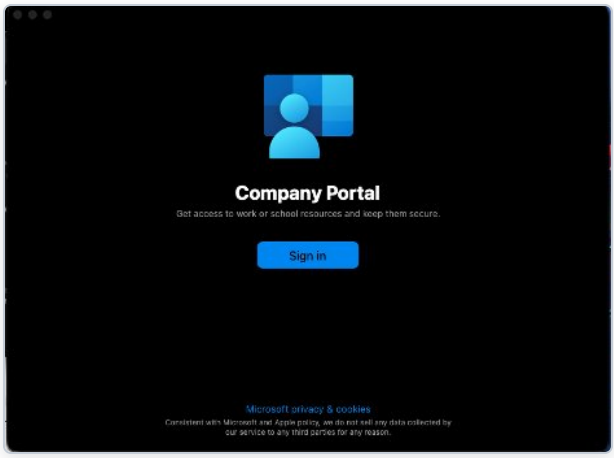



**Figure 10.** Company Portal sign-in screen.

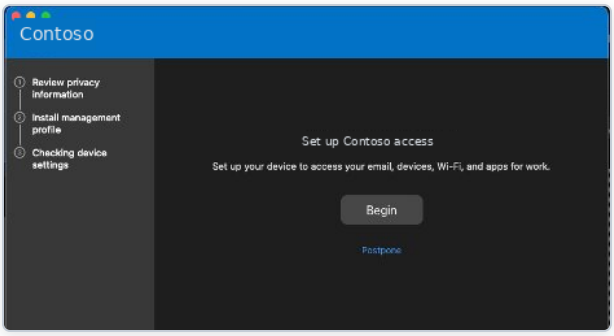



**Figure 11.** The setup screen shows the organization name configured in Intune branding (here "Contoso").

## STEP 6–7 Install and approve the management profile

1. On *Install management profile* select **Download profile**. macOS opens System Settings (the profile is downloaded but *not yet installed*).
2. In **System Settings › General › Device Management** select **Management Profile**, then **Install…**, and authenticate with the Mac's administrator password. (On macOS 13 and earlier the location is **System Settings › Privacy & Security › Profiles**.)
3. Return to Company Portal. It checks device settings against the organization's compliance policies. When *You're all set!* shows three green checkmarks, select **Done**.

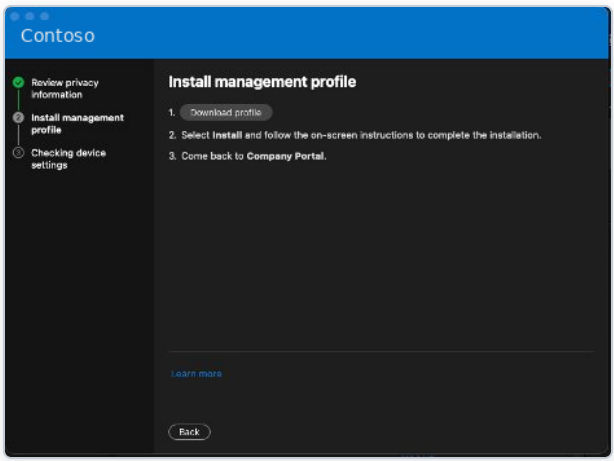



**Figure 12.** Install management profile: download, then approve in System Settings.

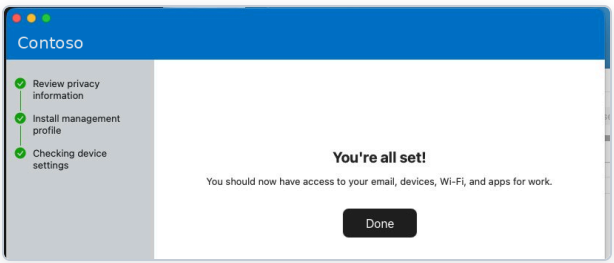



**Figure 13.** Enrollment complete: all three stages show a green check.

**KEY CONCEPT · WHY THE EXTRA TRIP TO SYSTEM SETTINGS?**

Since macOS 11 Apple does not allow any app, not even Company Portal, to install an MDM profile silently. A human must approve it in System Settings. That deliberate approval is precisely what makes the enrollment *user-approved MDM*, which later lets Intune deploy security tools such as Defender for Endpoint without further prompts.

**NOTE · HELP DESK SCRIPT**

If a user gets stuck, ask for the Mac's serial number (**Apple menu › About This Mac**) and a description of the screen they see. Do not ask them to send screenshots that include their email or other personal data unless necessary.

CHAPTER 4

# Method B · Admin Prerequisites in the Intune Admin Center

These are one-time (or yearly) tasks that every macOS enrollment path depends on.

## B1 Create (or renew) the Apple MDM push certificate

1. In **intune.microsoft.com** go to **Devices › Enrollment › Apple › Apple MDM Push certificate**.
2. Agree to the terms and select **Download your CSR** (certificate signing request).
3. Select **Create your MDM push Certificate**; the Apple Push Certificates Portal opens. Sign in with the **shared company Apple Account**.
4. Choose **Create a Certificate** (first time) or **Renew** next to the existing Intune certificate (yearly). Upload the CSR and download the resulting `.pem` file.
5. Back in Intune enter the Apple Account used, upload the `.pem` and select **Upload**. Confirm the new expiry date (one year ahead).

> **WATCH OUT · RENEW, NEVER RE-CREATE**
>
> In the Apple portal always use **Renew** on the existing certificate. A new certificate has a new topic identifier, and every Apple device enrolled under the old topic stops responding to Intune.

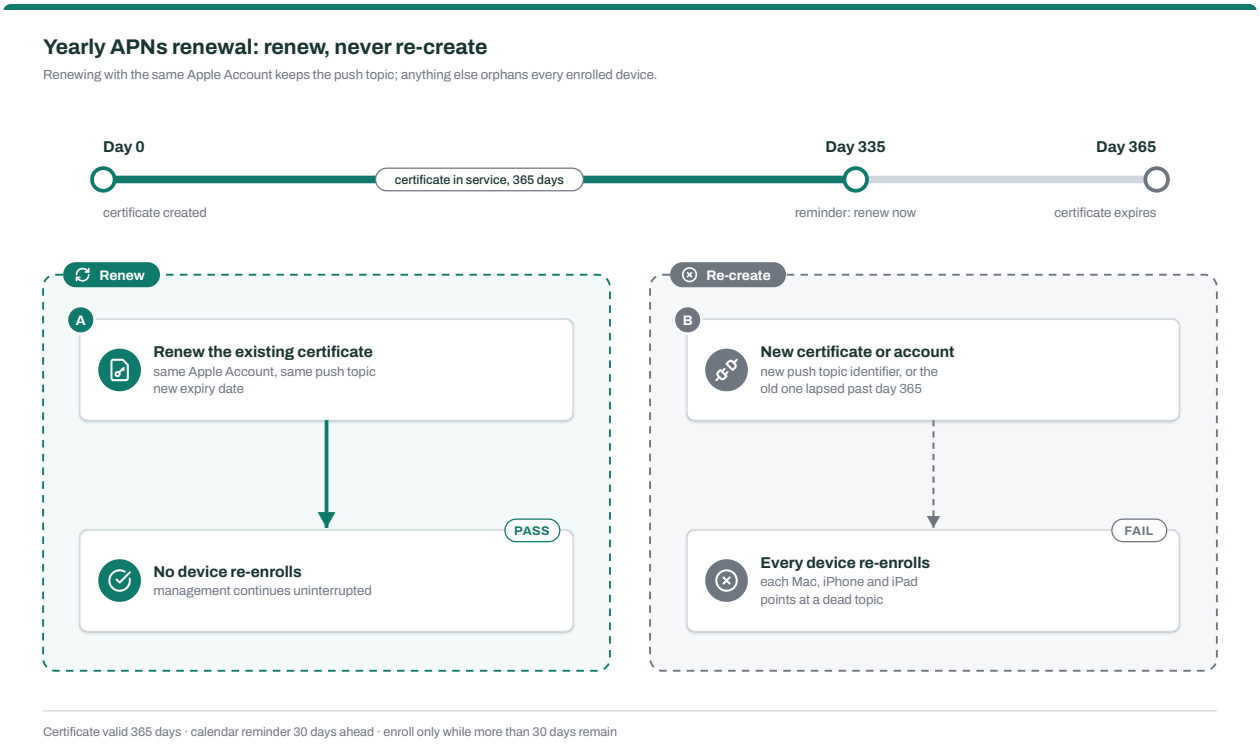



**Figure 14.** Always renew the existing certificate with the same Apple Account; a new certificate creates a new push topic and forces every Mac, iPhone and iPad to re-enroll.

## B2 Set enrollment restrictions and device limits

1. **Devices › Enrollment › Device platform restriction › macOS restrictions**: open the default policy (or create a policy for a specific group) and set *MDM* to **Allow**; set *Personally owned* to Allow or Block per your decision.
2. **Devices › Enrollment › Device limit restriction**: confirm the per-user limit (default 5, maximum 15). Users who hit the limit get a generic enrollment error.
3. Assign group-specific restrictions with priorities; the default policy applies to everyone else.

## B3 Register corporate Macs and set branding

1. **Devices › Enrollment › Corporate device identifiers › Add**: upload a CSV of Mac serial numbers so Company Portal enrollments of those Macs are marked *Corporate* (and are allowed even when personal Macs are blocked).

2 **Tenant administration › Customization:** set the organization name, support contact and privacy statement URL that Company Portal displays.

CORPORATE-IDENTIFIERS.CSV (EXAMPLE, NO HEADER ROW) TEXT

```
C02EXAMPLE01,MacBook Pro - Finance
C02EXAMPLE02,MacBook Air - Lab
```

CHAPTER 5

# Method C · Automated Device Enrollment with Apple Business Manager

ADE is the gold standard for corporate Macs: the user unboxes the Mac, connects to Wi-Fi, signs in with their work account in Setup Assistant, and the Mac arrives at the desktop already supervised, enrolled and receiving policy. The setup links ABM and Intune with a **token** (a signed file that lets Intune read the device list ABM assigns to it).

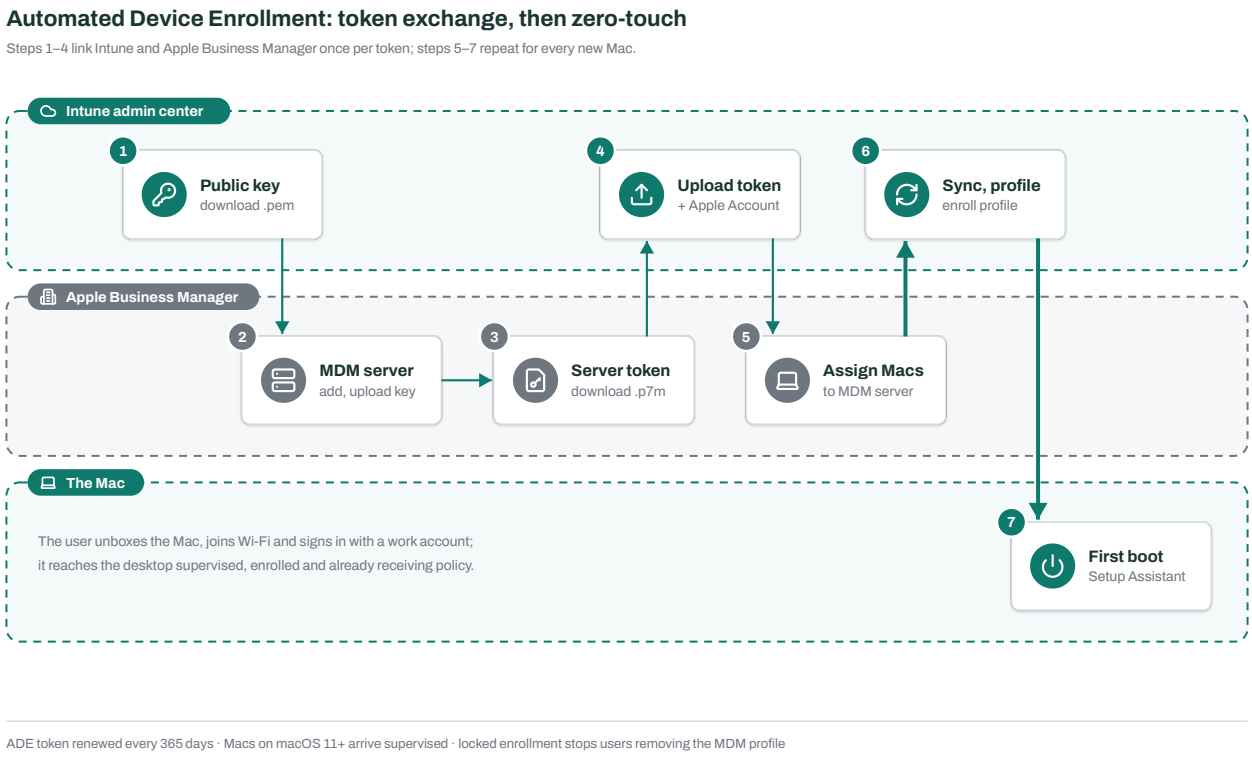



**Figure 15.** The token exchange is a once-a-year administrative task, after which every new Mac enrolls itself during Setup Assistant with no IT touch.

## C1 Create the enrollment program token

1. In Intune: **Devices › Enrollment › Apple › Enrollment program tokens › Add**. Agree to the terms and **Download your public key** (`.pem`).
2. In Apple Business Manager (**business.apple.com**): select your name › **Preferences › Your MDM Servers › Add** (labels vary slightly between ABM releases), name the server (for example *Contoso Intune*), upload the public key and save. Download the **server token** (`.p7m`).

3. Back in Intune enter the Apple Account used for ABM, upload the `.p7m` and create the token. Note its one-year expiry.
4. In ABM set *Default Device Assignment* for Mac to the new MDM server, or assign existing Macs by serial number or order number (**Devices › select › Edit MDM Server**).

## C2 Create and assign the macOS enrollment profile

1. Open the token, select **Sync** and confirm the Macs appear under *Devices*.
2. **Profiles › Create profile › macOS**. Recommended settings for an assigned corporate Mac:
   - *User affinity*: Enroll with User Affinity; *Authentication method*: Setup Assistant with modern authentication.
   - *Locked enrollment*: Yes (users cannot remove the MDM profile).
   - *Await final configuration*: Yes (holds the user in Setup Assistant until critical policies arrive).
   - *Setup Assistant*: hide screens the organization does not need (for example Apple Pay, Siri, Screen Time).
3. Save, then **Set default profile** for the token or assign the profile to specific Macs.
4. Assign the Company Portal app (Required) to macOS users so modern-authentication enrollments complete user registration.

**GOOD PRACTICE · RE-ENROLL EXISTING MACS INTO ADE**

A Mac already in use can be brought into ADE: add it to ABM (Apple Configurator for iPhone can add Macs), assign it to the Intune MDM server, then erase it, or on a Company Portal Mac run `sudo profiles renew -type enrollment` to start ADE enrollment (supervision still requires an erase on most releases).

CHAPTER 6

# Method D · Graph PowerShell Readiness Template

Enrollment fails silently when a certificate or token expires, so the most valuable automation is a scheduled readiness check. The template below uses the **Microsoft Graph PowerShell SDK** to report the APNs certificate and ADE token expiry, the macOS enrollment restrictions and every enrolled Mac, writes CSV and JSON evidence, and can optionally trigger an ADE token sync (the only write action, guarded by `-WhatIf`).

GET-MACENROLLMENTREADINESS.PS1 · BASE TEMPLATE POWERSHELL

```powershell
<#
.SYNOPSIS  Report macOS enrollment readiness in Intune: APNs certificate, ADE tokens,
           enrollment restrictions and enrolled Macs. Optional: sync ADE tokens.
.NOTES     Modules: Microsoft.Graph.Authentication, Microsoft.Graph.DeviceManagement
           Roles  : Read Only Operator (read); Enrollment programs role (-SyncAdeTokens)
           BETA   : depOnboardingSettings and per-platform restrictions use /beta (may change).
#>
[CmdletBinding(SupportsShouldProcess)]
param(
  [int]    $WarnDays     = 30,                          # flag certs/tokens expiring sooner
  [int]    $StaleDays    = 14,                          # flag Macs not checked in since
  [switch] $SyncAdeTokens,                              # write action: sync tokens with Apple
  [string] $OutputFolder = ".\mac-enrollment-$(Get-Date -Format yyyyMMdd)"
)
$ErrorActionPreference = 'Stop'
$v1 = 'https://graph.microsoft.com/v1.0'; $beta = 'https://graph.microsoft.com/beta'
$svc = if ($SyncAdeTokens) { 'DeviceManagementServiceConfig.ReadWrite.All' }
       else                { 'DeviceManagementServiceConfig.Read.All' }
Connect-MgGraph -NoWelcome -Scopes $svc, 'DeviceManagementManagedDevices.Read.All'
New-Item -ItemType Directory -Path $OutputFolder -Force | Out-Null
$now = Get-Date
$findings = [System.Collections.Generic.List[object]]::new()
function Add-Finding($Area, $Item, $Status, $Detail) {
  $findings.Add([pscustomobject]@{ Area = $Area; Item = $Item; Status = $Status; Detail = $Detail })
}
function Get-Status([datetime]$Expiry) {
  $d = [int]($Expiry - $now).TotalDays
  if ($d -lt 0) { 'EXPIRED' } elseif ($d -le $WarnDays) { 'WARN' } else { 'OK' }
}

# ---- Stage 1: APNs certificate (v1.0) -------------------------------------
$apns = Invoke-MgGraphRequest GET "$v1/deviceManagement/applePushNotificationCertificate"
$exp  = [datetime]$apns.expirationDateTime
Add-Finding 'APNs' $apns.appleIdentifier (Get-Status $exp) "Expires $($exp.ToString('u'))"

# ---- Stage 2: ADE (enrollment program) tokens (BETA) ----------------------
$tokens = (Invoke-MgGraphRequest GET "$beta/deviceManagement/depOnboardingSettings").value
foreach ($t in $tokens) {
  $texp = [datetime]$t.tokenExpirationDateTime
  $detail = "Apple ID $($t.appleIdentifier); expires $($texp.ToString('u'))" +
            "; last sync $($t.lastSuccessfulSyncDateTime)"
  Add-Finding 'ADE token' $t.tokenName (Get-Status $texp) $detail
  if ($SyncAdeTokens -and $PSCmdlet.ShouldProcess($t.tokenName, 'Sync with Apple')) {
    $uri = "$beta/deviceManagement/depOnboardingSettings/$($t.id)/syncWithAppleDeviceEnrollmentProgram"
    Invoke-MgGraphRequest POST $uri | Out-Null
  }
```

```
}
if (-not $tokens) { Add-Finding 'ADE token' '(none)' 'INFO' 'No ADE token - Company Portal only' }

# ---- Stage 3: enrollment restrictions (BETA, both policy generations) -------
$cfgs = (Invoke-MgGraphRequest GET "$beta/deviceManagement/deviceEnrollmentConfigurations").value
foreach ($c in $cfgs) {
  $type = $c.'@odata.type'
  if ($type -like '*PlatformRestrictionsConfiguration') {          # legacy default policy
    $r = $c.macOSRestriction }
  elseif ($type -like '*PlatformRestrictionConfiguration' -and $c.platformType -eq 'macOS') {
    $r = $c.platformRestriction }                                   # per-platform policy
  elseif ($type -like '*LimitConfiguration') {
    Add-Finding 'Device limit' $c.displayName 'INFO' "Limit $($c.limit) devices per user"; continue }
  else { continue }
  $st = if ($r.platformBlocked) { 'BLOCKED' } else { 'OK' }
  $detail = "macOS blocked=$($r.platformBlocked); " +
            "personal blocked=$($r.personalDeviceEnrollmentBlocked)"
  Add-Finding 'Restriction' $c.displayName $st $detail
}

# ---- Stage 4: enrolled Macs (v1.0) ------------------------------------------
$macs = Get-MgDeviceManagementManagedDevice -Filter "operatingSystem eq 'macOS'" -All |
  Select-Object DeviceName, UserPrincipalName, SerialNumber, OSVersion, ManagedDeviceOwnerType,
                DeviceEnrollmentType, IsSupervised, EnrolledDateTime, LastSyncDateTime, ComplianceState
$stale = @($macs | Where-Object { $_.LastSyncDateTime -lt $now.AddDays(-$StaleDays) })
$detail = "$(@($macs).Count) Macs; $($stale.Count) not seen in $StaleDays days"
Add-Finding 'Devices' 'macOS' 'INFO' $detail
$macs | Group-Object DeviceEnrollmentType | ForEach-Object {
  Add-Finding 'Devices' $_.Name 'INFO' "$($_.Count) Macs" }   # appleBulkWith(out)User = ADE

# ---- Evidence ---------------------------------------------------------------
$macs     | Export-Csv "$OutputFolder\macos-devices.csv" -NoTypeInformation   # contains UPNs/serials
$findings | Export-Csv "$OutputFolder\readiness.csv"     -NoTypeInformation
$findings | ConvertTo-Json | Out-File "$OutputFolder\readiness.json" -Encoding utf8
$findings | Format-Table Area, Item, Status, Detail -AutoSize -Wrap
if ($findings.Status -match 'EXPIRED|WARN|BLOCKED') {
  Write-Warning 'Action needed - see readiness.csv' }
```

**EXAMPLE · RUNNING THE TEMPLATE**

INVOCATION — POWERSHELL

```
.\Get-MacEnrollmentReadiness.ps1 -WarnDays 45                  # read-only report
.\Get-MacEnrollmentReadiness.ps1 -SyncAdeTokens -WhatIf        # preview the token sync
.\Get-MacEnrollmentReadiness.ps1 -SyncAdeTokens                # sync after a new ABM purchase
```

Schedule the read-only form weekly (for example in an Azure Automation runbook using a managed identity with the read scopes as *application* permissions) and alert on any `WARN` or `EXPIRED` row. The device CSV contains user names and serial numbers; store it with the same care as other personal data.

**NOTE · BETA ENDPOINTS**

ADE token and per-platform restriction data are only exposed on Graph `beta`. Beta resources can change without notice; pin the script's behavior with a test run after Graph updates and prefer the admin center for one-off changes.

## 6.1 Method E · Direct enrollment (no user affinity)

Use direct enrollment for shared Macs (labs, kiosks, conference rooms) that will never have a user signing in to Company Portal, and that cannot go through ADE.

1. In Intune create an Apple Configurator enrollment profile for the Mac without user affinity (**Devices › Enrollment › Apple › Apple Configurator › Profiles**; the location can vary by console release).
2. Select the profile and **Export profile**; download the `.mobileconfig`.
3. Copy the profile to the Mac (or push it with Apple Configurator), open it, and approve it in **System Settings › General › Device Management** with administrator credentials.
4. Confirm the Mac appears in Intune with no primary user; target it with device-based (not user-based) policies.

A **device enrollment manager (DEM)** account is a related option: one service account enrolls many Macs through Company Portal. DEM devices cannot use conditional access per user and are best avoided where ADE is possible.

CHAPTER 7

# Verification and Troubleshooting

## 7.1 Verify on the Mac

TERMINAL ON THE MAC — BASH

```
sudo profiles status -type enrollment
#   Enrolled via DEP: No|Yes          (Yes = Automated Device Enrollment)
#   MDM enrollment: Yes (User Approved)
sudo profiles show -type enrollment   # ADE Macs: shows the configuration Apple holds for this serial
sudo profiles list                    # MDM profile + configuration profiles installed by Intune
```

- [ ] **System Settings › General › Device Management** lists the organization's *Management Profile* as verified.
- [ ] Company Portal › **Devices** shows the Mac as compliant (or explains what is missing).

## 7.2 Verify in Intune

- [ ] **Devices › macOS** lists the Mac with the correct primary user, *Ownership* (Corporate/Personal) and a recent *Last check-in*.
- [ ] The device **Properties** show the expected enrollment type (for ADE, "Apple automated device enrollment") and *Supervised: Yes* for ADE Macs.
- [ ] The readiness template reports no `WARN` / `EXPIRED` rows and the Mac in the device CSV.

## 7.3 Troubleshooting

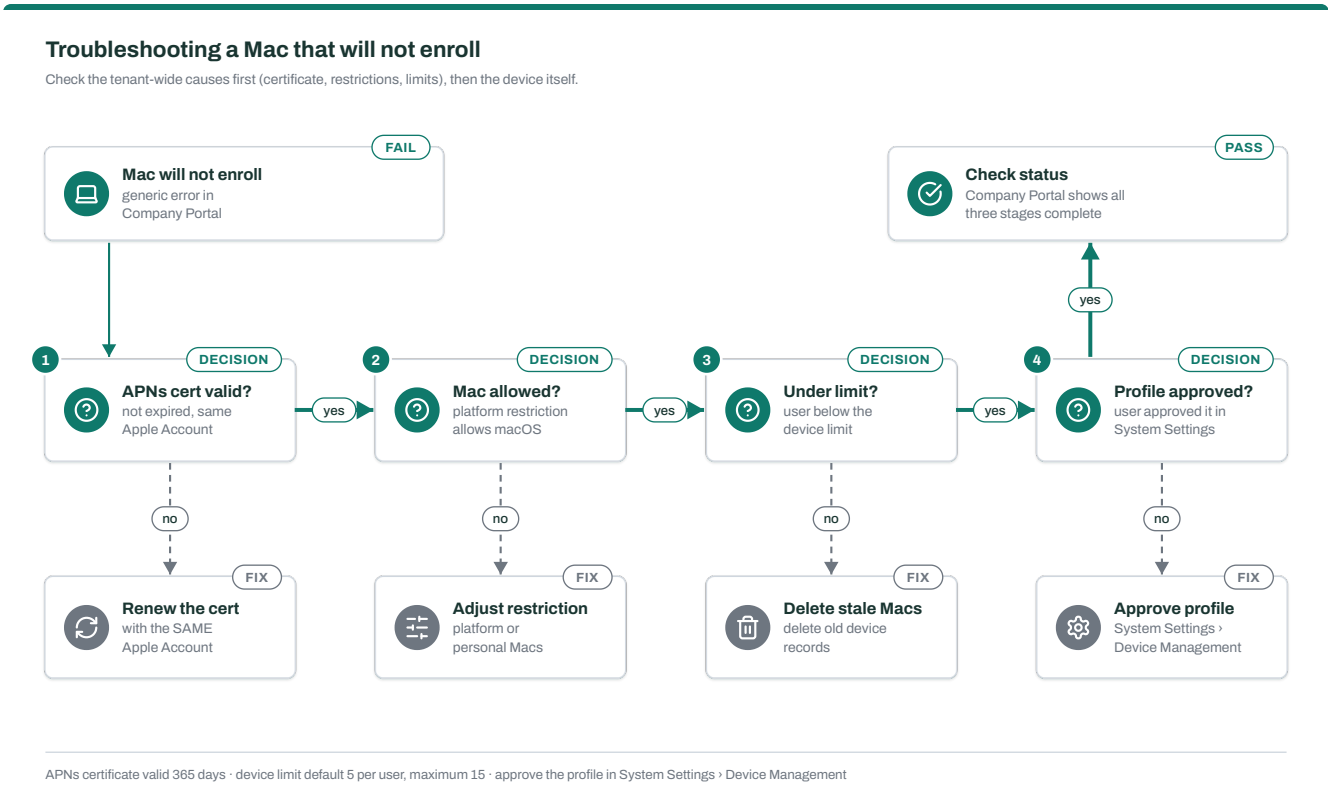



**Figure 16.** Tenant-wide causes such as the certificate, restrictions and limits break every enrollment at once, so rule them out before debugging a single Mac.

| Symptom | Likely cause and fix |
|---|---|
| **"Couldn't add your device" or enrollment blocked** | macOS or personal enrollment blocked by platform restriction, or the user has reached the device limit. Adjust restriction or remove stale records. |
| **All Apple devices stop checking in** | APNs certificate expired or was re-created under a different Apple Account. Renew with the original account; if lost, devices must re-enroll. |
| **Profile downloaded but Company Portal waits forever** | The user did not approve it in System Settings › Device Management. Walk them through Step 6. |
| **ADE Mac skips enrollment in Setup Assistant** | Mac not assigned to the Intune MDM server in ABM, token not synced, or no profile assigned. Assign, sync, then erase or run `sudo profiles renew -type enrollment`. |
| **ADE token shows an error** | Token expired or the ABM terms changed. Renew the token in ABM, upload to Intune, sync. |

| | |
|---|---|
| **Corporate Mac shows as Personal** | Serial not in corporate identifiers before enrollment. Change ownership in device Properties and add the serial for future enrollments. |
| **Mac enrolled but no policies arrive** | Network blocks APNs (TCP 5223 / 443 to Apple). Test from the Mac; exempt Apple and Intune endpoints from TLS inspection. |

## 7.4 Rollback and unenrollment

To remove a Mac from management use the `Retire` device action (removes company data and profiles, keeps personal data) or `Wipe` (erase) for corporate ADE Macs being reassigned. Users can remove an unlocked Company Portal enrollment themselves in Company Portal or System Settings. A locked ADE profile can only be removed by Intune or by erasing the Mac. For a failed tenant-level change (for example a restriction that blocked everyone), revert the policy; already enrolled Macs are not affected by restriction changes.

CHAPTER 8

# Summary, Glossary and Review

## 8.1 Chapter summary

- Every path installs an MDM enrollment profile; the APNs certificate lets Intune wake the Mac and must be renewed yearly with the same Apple Account.
- Company Portal gives a user-approved, unsupervised, removable enrollment; ADE gives a supervised, locked, zero-touch one; direct enrollment covers shared Macs.
- Restrictions, device limits, corporate identifiers and branding are set once and shape every enrollment.
- A scheduled Graph readiness report catches expiring certificates and tokens before users notice.

**Check your understanding**

1. Why does Intune need Apple's push service to manage a Mac, and what happens when the APNs certificate expires?
2. What is the practical difference between a user-approved and a supervised Mac?
3. A user downloaded the profile but Company Portal never finishes. What did they miss?
4. Which ABM and Intune objects must exist before a new Mac enrolls itself in Setup Assistant?
5. When is direct enrollment preferable to Company Portal?

## 8.2 Glossary

| | |
|---|---|
| **APNs** | Apple Push Notification service; wakes devices so they check in with their MDM server. |
| **ABM** | Apple Business Manager; Apple's portal for device ownership, MDM server assignment and Managed Apple Accounts. |
| **ADE** | Automated Device Enrollment (formerly DEP); enrollment during Setup Assistant for ABM-registered devices. |
| **Supervision** | Higher management trust level granted to ADE-enrolled devices. |
| **User-approved MDM** | Enrollment the user explicitly approved, or any ADE enrollment. |
| **User affinity** | Association of a device with a primary user in Intune. |
| **Corporate identifier** | Pre-registered serial that marks a Mac as corporate. |